



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ASSESSMENT TOOL 3 – CO1

EXPLORATORY DATA ANALYSIS

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Course Code: DSA0402

Course Name: Fundamentals of Data Science

QUESTIONS

Question 1 – Education Analytics Dataset

A college wants to analyze student performance based on attendance, study hours, internal marks, and final result. Perform EDA to identify the factors affecting student results.

Question 2 – Healthcare Analytics Dataset

A hospital wants to analyze patient health data to identify patterns related to disease status using age, sugar level, blood pressure, and BMI.

Question 3 – Retail Sales Dataset

A supermarket wants to analyze sales data to identify product categories and pricing patterns that generate the most revenue.

Question 4 – Banking Loan Approval Dataset

A bank wants to analyze customer and loan data to identify factors influencing loan approval decisions.

Question 5 – Agriculture Crop Yield Dataset

An agricultural department wants to analyze rainfall, temperature, fertilizer usage, and soil type to identify factors influencing crop yield.

QUESTION 1 – EDUCATION ANALYTICS DATASET

1.1 Dataset Description

The Education Analytics dataset contains 20 records and 5 features. The features are Student_ID, Attendance, Study_Hours, Internal_Marks, and Result. Student_ID is an object variable, Attendance, Study_Hours and Internal_Marks are numerical variables, and Result is a categorical target variable.

Feature	Data Type	Role
Student_ID	object	Unique identifier
Attendance	int64	Numerical predictor (%)
Study_Hours	int64	Numerical predictor (hrs/day)
Internal_Marks	int64	Numerical predictor (/50)
Result	object	Target: Pass / Fail

1.2 Purpose and Significance

A college wants to understand which factors, such as attendance, daily study hours, and internal assessment marks, most influence whether a student passes or fails. This analysis can support early-warning systems and mentoring interventions.

1.3 Data Cleaning and Preprocessing

- No missing values were found.
- No duplicate records were found.
- Result labels were checked and only Pass and Fail were present.
- No outliers were detected using the IQR rule.
- Therefore, no imputation, duplicate removal, or outlier capping was required.

1.4 Descriptive Statistics

Feature	Mean	Maximum	Minimum	Std. Dev
Attendance	70.15	95	45	15.94
Study Hours	3.35	6	1	1.42
Internal Marks	32.95	48	18	9.09

These values are reported in the assessment document.

1.5 Categorical Variable Analysis

Variable	Category	Count	Percentage
Result	Pass	11	55%
Result	Fail	9	45%

1.6 Visualizations

Figure 1.1 – Education Overview

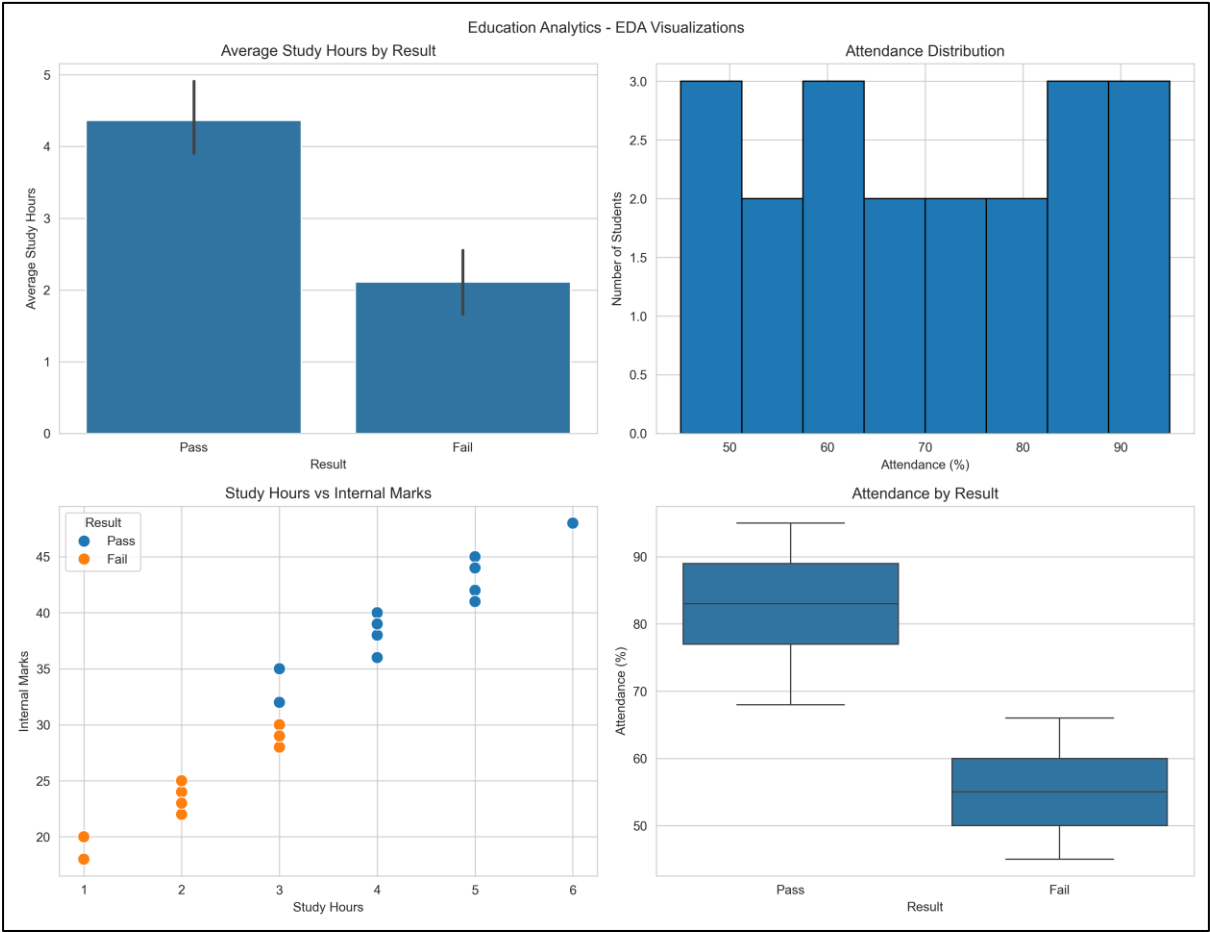


Figure 1.2 – Education Correlation Heatmap

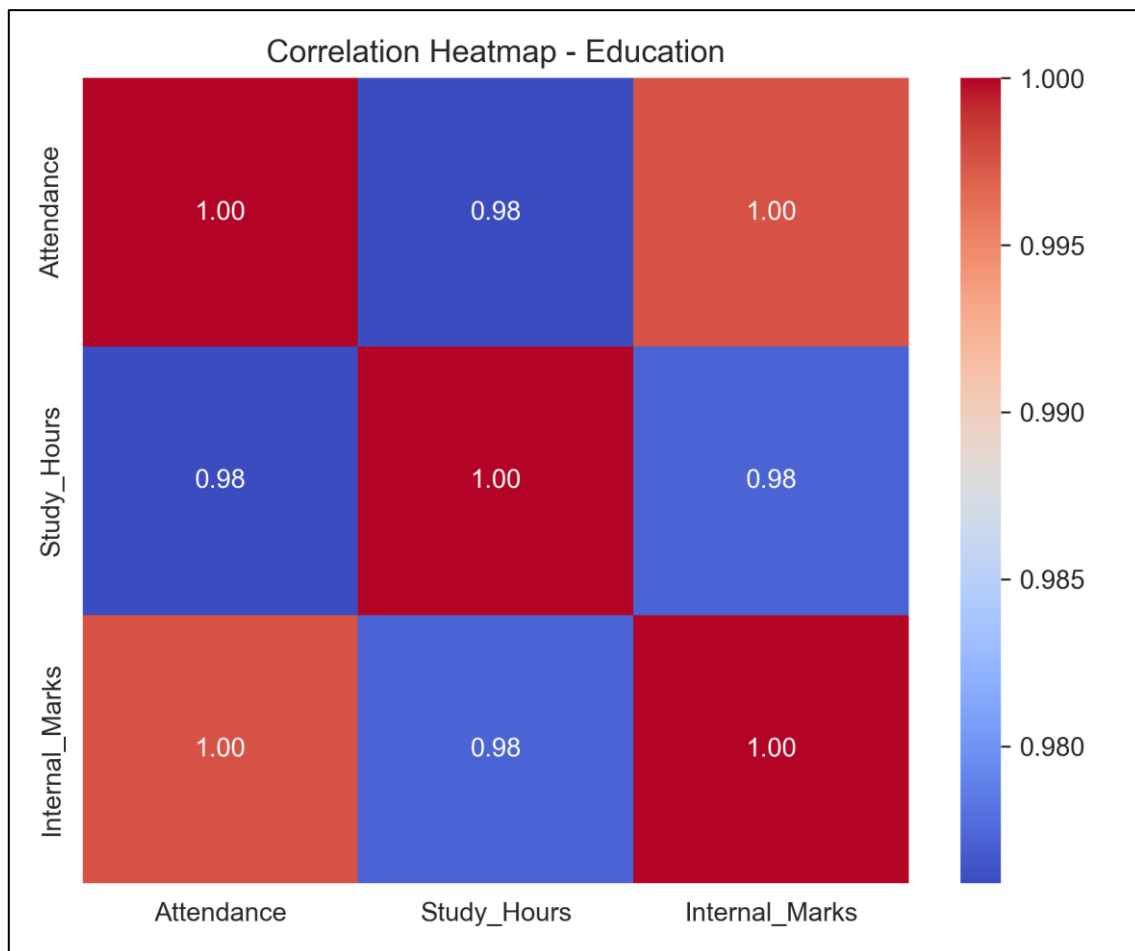


Figure 1.3 – Education Line Chart

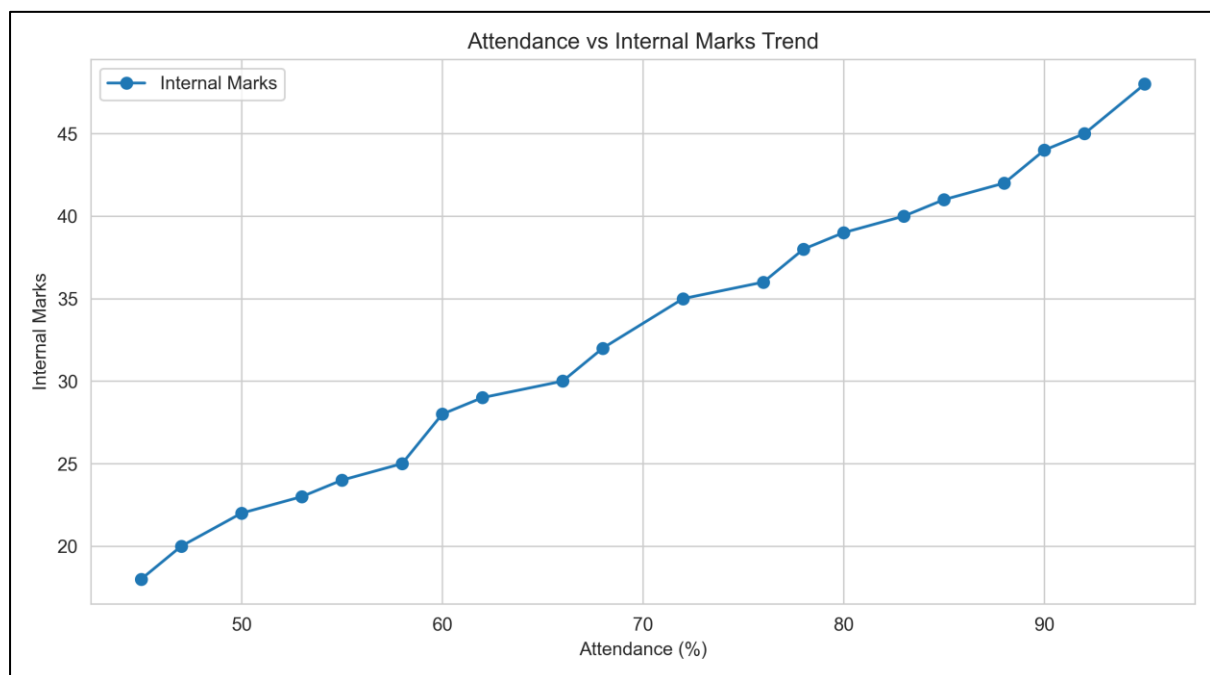
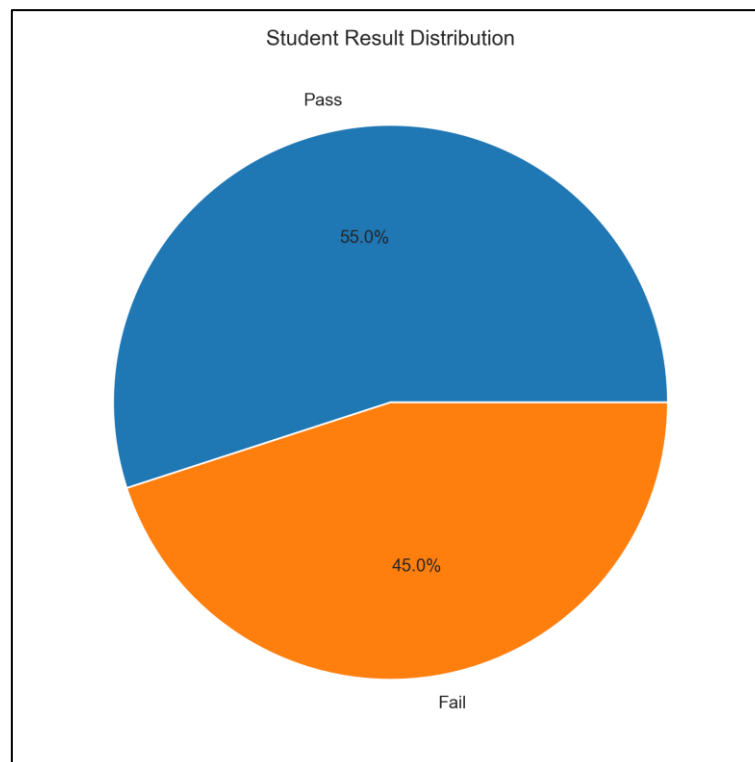


Figure 1.4 – Education Pie Chart



The assessment specifies that visualizations should have explicit titles, axis labels and legends where applicable.

1.7 Trends, Distributions and Relationships

Attendance and Internal Marks have an almost perfect positive linear relationship with $r = 0.998$. Study Hours also has a strong positive relationship with Attendance ($r = 0.976$) and Internal Marks ($r = 0.977$). Attendance ranges from 45% to 95%, with a visible separation between Pass and Fail outcomes around the 60–65% range.

1.8 Key Findings and Insights

1. Passing students show higher average attendance and study hours than failing students.
2. Internal marks increase almost linearly with study hours.
3. Attendance below approximately 60% is strongly associated with a Fail result.
4. Attendance, study hours and internal marks are all positively correlated.
5. The dataset is complete, unique and free from detected outliers.

1.9 Decision-Making Impact

Academic advisors can use the approximately 60% attendance threshold to identify students who may need mentoring. Structured study targets of around 4 or more hours per day can also be encouraged to improve internal marks and pass rates.

QUESTION 2 – HEALTHCARE ANALYTICS DATASET

2.1 Dataset Description

The Healthcare Analytics dataset contains 20 records and 6 features. It includes Patient_ID, Age, Sugar_Level, BP, BMI and Disease_Status.

Feature	Data Type	Role
Patient_ID	object	Unique identifier
Age	int64	Numerical predictor
Sugar_Level	int64	Numerical predictor
BP	int64	Numerical predictor
BMI	int64	Numerical predictor
Disease_Status	object	Target: Yes / No

2.2 Purpose and Significance

A hospital wants to identify which combination of age, sugar level, blood pressure and BMI most reliably indicates disease presence so that patients can be prioritized for preventive screening.

2.3 Data Cleaning and Preprocessing

- 0 missing values.
- 0 duplicate rows.
- Disease_Status contains consistent Yes and No labels.
- 0 IQR outliers were detected.

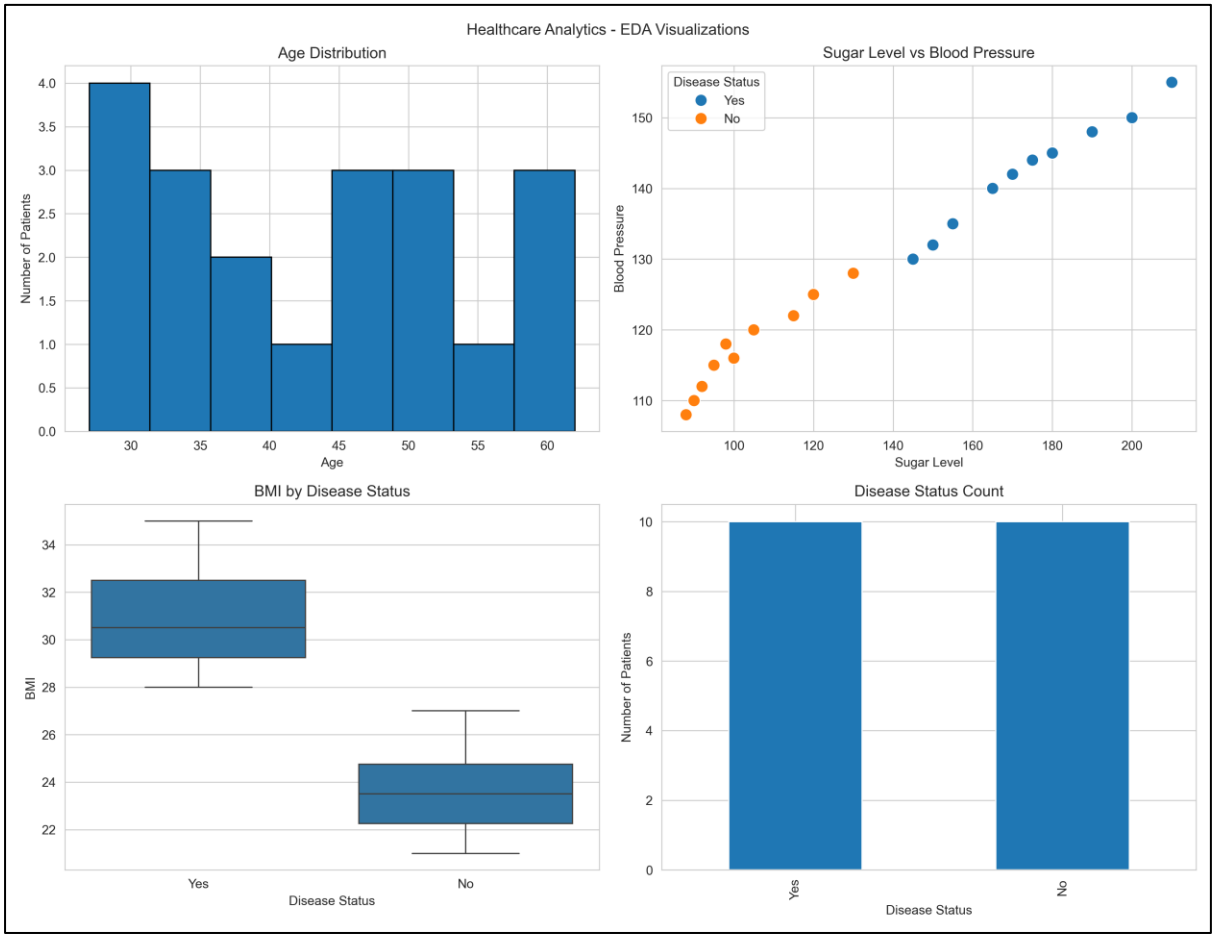
2.4 Descriptive Statistics

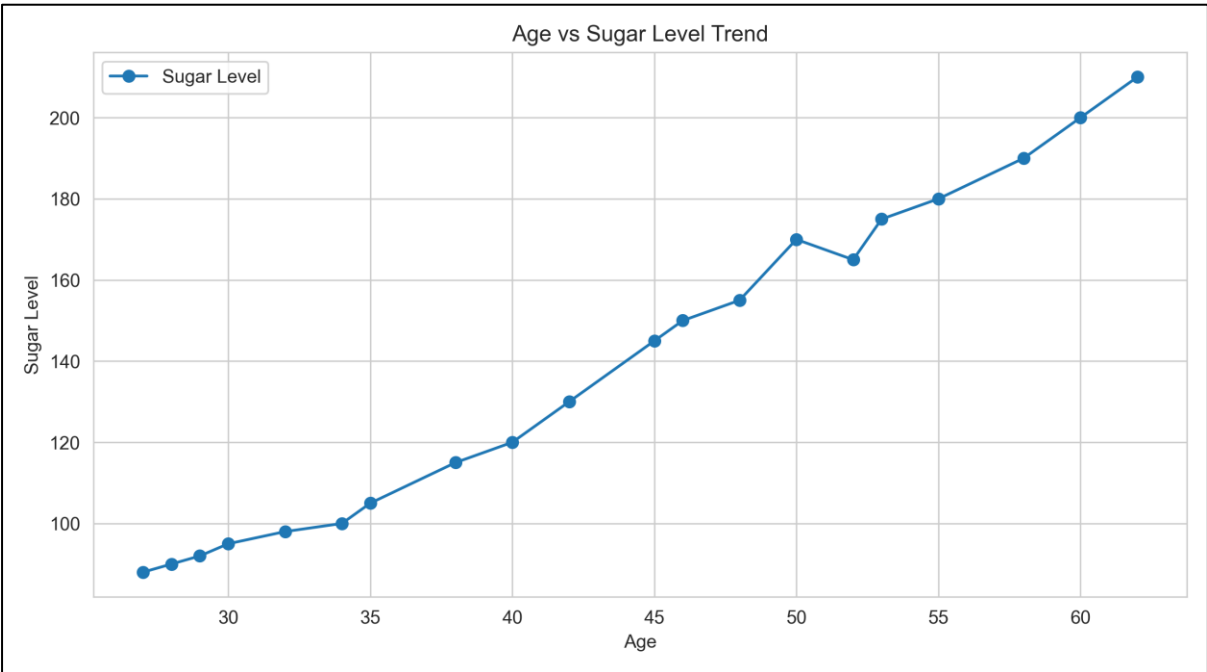
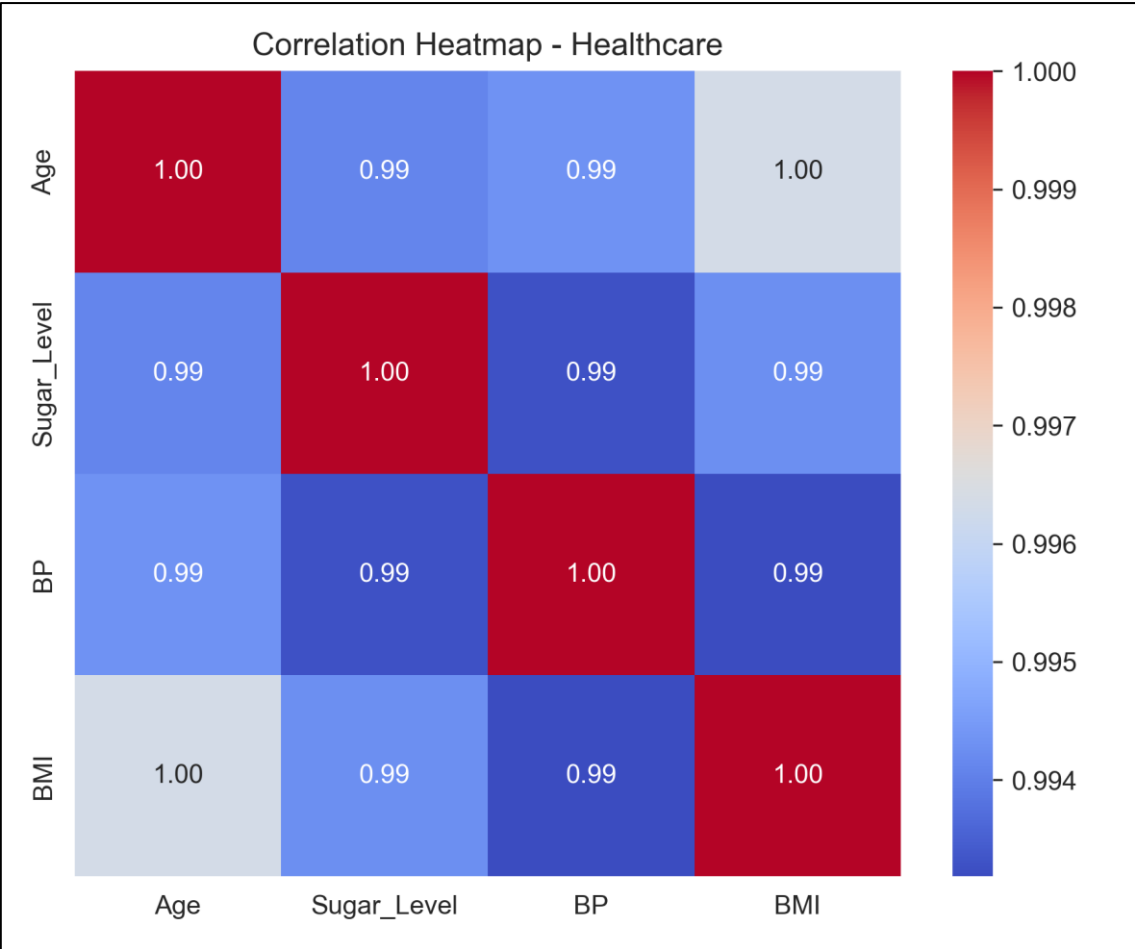
Feature	Mean	Maximum	Minimum	Std. Dev
Age	43.20	62	27	11.32
Sugar Level	138.65	210	88	40.33
BP	129.75	155	108	14.52
BMI	27.30	35	21	4.26

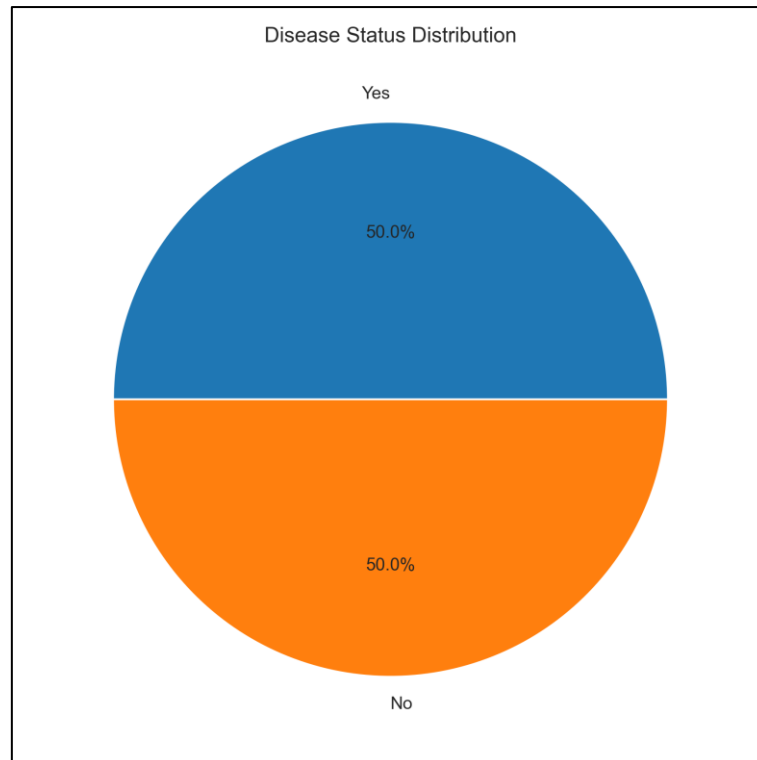
2.5 Categorical Variable Analysis

Category	Count	Percentage
Yes	10	50%
No	10	50%

2.6 Visualizations







2.7 Trends, Distributions and Relationships

The numerical variables show very strong positive relationships, with pairwise correlations between 0.993 and 0.996. BMI provides the clearest visual separation between disease-positive and disease-negative groups.

2.8 Key Findings and Insights

1. Disease-positive patients show higher sugar level, BP, BMI and age.
2. Sugar level and BP have a strong positive relationship.
3. BMI provides a clear separation between the two disease-status groups.
4. The dataset is balanced with 10 Yes and 10 No records.
5. No missing values, duplicates or IQR outliers were detected.

2.9 Decision-Making Impact

BMI and sugar-level thresholds can be used as preliminary screening indicators to identify patients who may require further diagnostic examination.

QUESTION 3 – RETAIL SALES DATASET

3.1 Dataset Description

The Retail Sales dataset contains **20 records and 6 features**: Product_ID, Category, Price, Quantity_Sold, Discount and Revenue.

Feature	Data Type	Role
Product_ID	object	Unique identifier
Category	object	Categorical
Price	int64	Numerical predictor
Quantity_Sold	int64	Numerical predictor
Discount	int64	Numerical predictor (%)
Revenue	int64	Numerical target

3.2 Purpose and Significance

A supermarket wants to identify which product categories and pricing patterns generate the most revenue. The analysis can support stocking, pricing and promotion decisions.

3.3 Data Cleaning and Preprocessing

- 0 missing values.
- 0 duplicate rows.
- All category labels were consistent.
- 0 IQR outliers were detected.

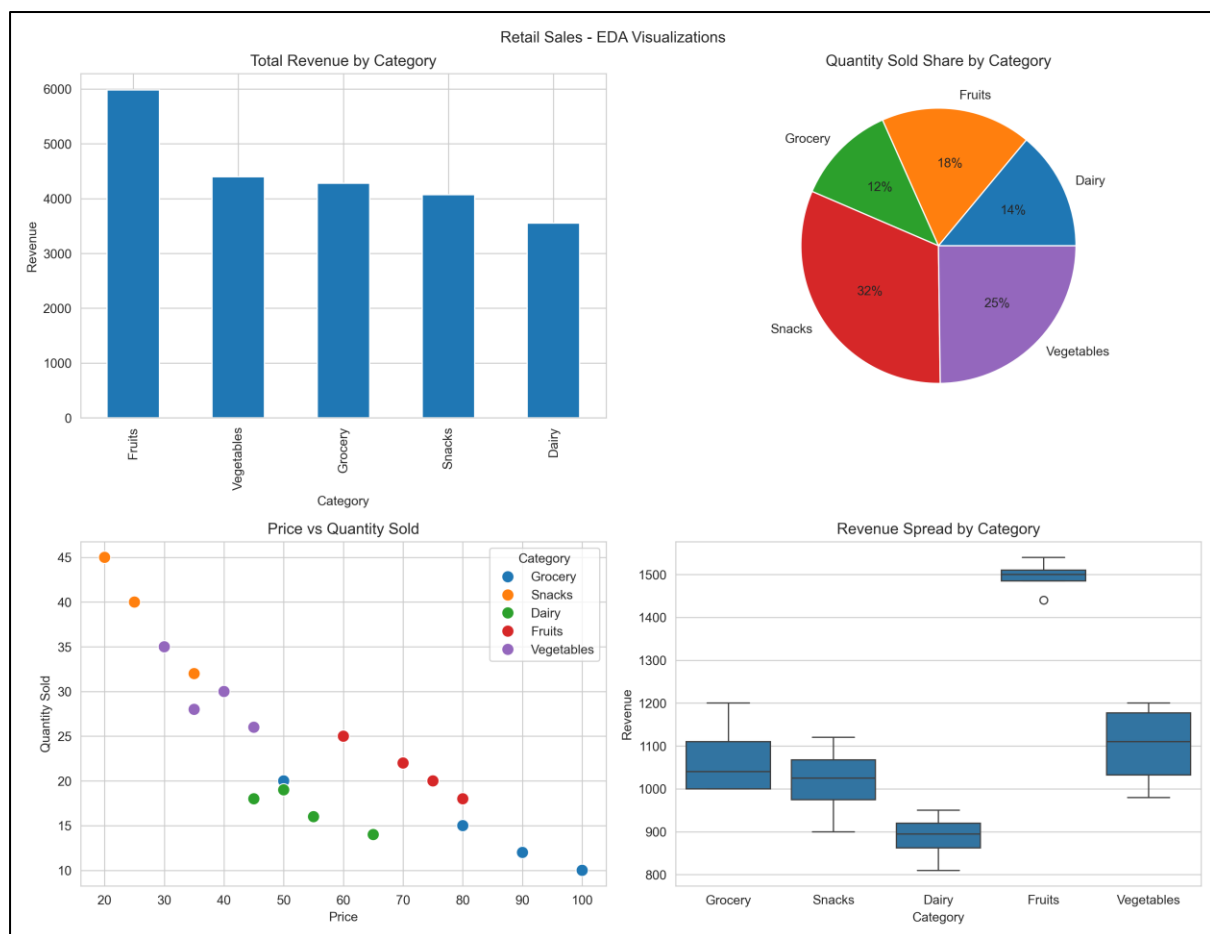
3.4 Descriptive Statistics

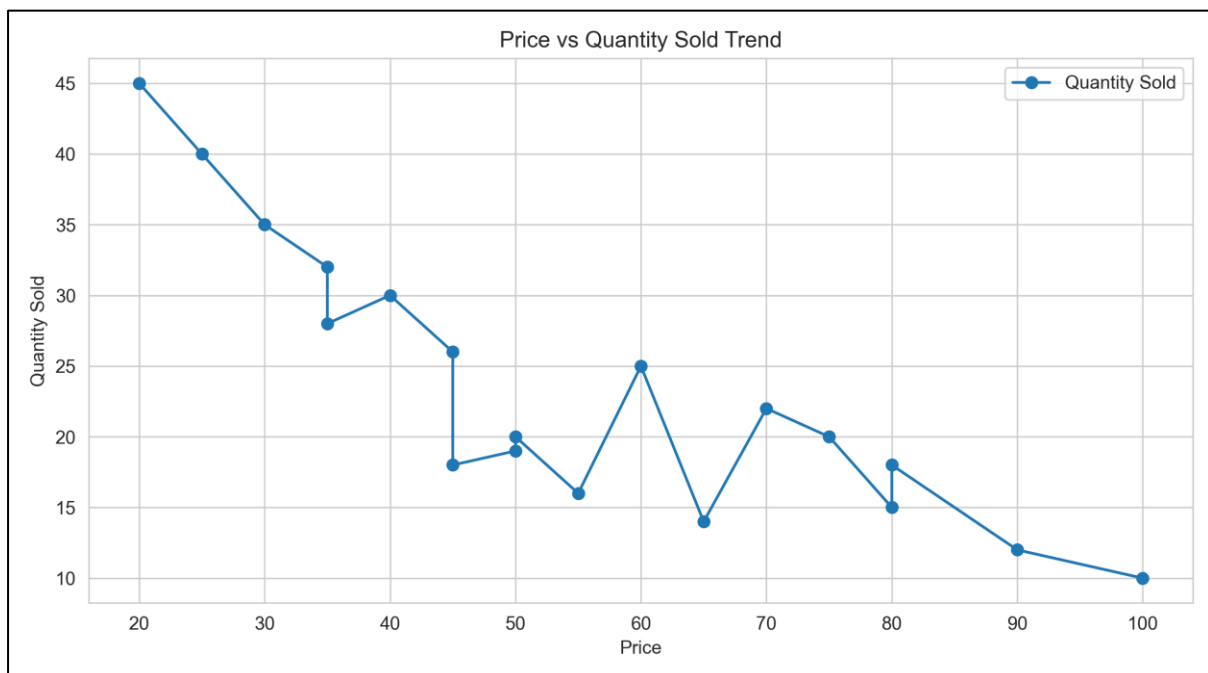
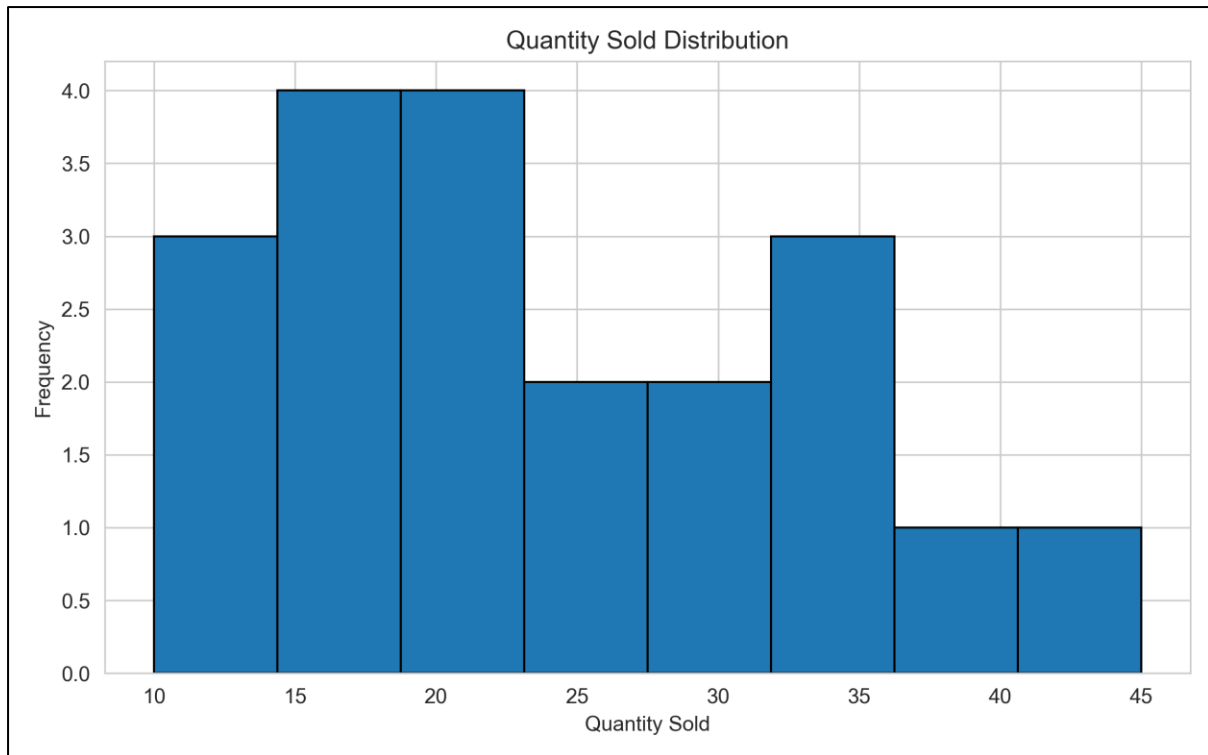
Feature	Mean	Maximum	Minimum	Std. Dev
Price	54.00	100	20	22.80
Quantity Sold	24.00	45	10	9.67
Discount	4.00	7	2	1.56
Revenue	1114.00	1540	810	221.42

3.5 Categorical Variable Analysis

Category	Count	Percentage
Grocery	4	20%
Snacks	4	20%
Dairy	4	20%
Fruits	4	20%
Vegetables	4	20%

3.6 Visualizations





3.7 Trends, Distributions and Relationships

Price and Quantity Sold have a strong negative relationship ($r = -0.868$), showing that quantity sold decreases as price increases. Revenue is not equally distributed across categories, with Fruits and Vegetables performing strongly due to higher price points.

3.8 Key Findings and Insights

1. Fruits generate the highest total revenue.
2. Price and quantity sold show a clear inverse relationship.
3. Grocery items have the highest discounts but comparatively lower quantity sold.
4. Dairy has the tightest revenue spread.
5. No missing values, duplicates or IQR outliers were found.

3.9 Decision-Making Impact

The results can help prioritize shelf space and stock levels for Fruits and Vegetables while using discounts strategically for Grocery products when the objective is increasing sales volume.

QUESTION 4 – BANKING LOAN APPROVAL DATASET

4.1 Dataset Description

The Banking Loan Approval dataset contains 20 records and 6 features. It includes Customer_ID, Income, Credit_Score, Loan_Amount, Employment_Type and Loan_Status.

Feature	Data Type	Role
Customer_ID	object	Unique identifier
Income	int64	Numerical predictor
Credit_Score	int64	Numerical predictor
Loan_Amount	int64	Numerical predictor
Employment_Type	object	Categorical
Loan_Status	object	Target

4.2 Purpose and Significance

A bank wants to understand which applicant attributes drive loan approval decisions to support fair and consistent underwriting policies.

4.3 Data Cleaning and Preprocessing

- 0 missing values.
- 0 duplicate rows.
- Employment_Type and Loan_Status labels were consistent.
- 0 IQR outliers were detected.

4.4 Descriptive Statistics

Feature	Mean	Maximum	Minimum	Std. Dev
Income	43000	75000	22000	17923.82
Credit Score	664.00	790	555	85.53
Loan Amount	223250	400000	120000	83733.21

4.5 Categorical Variable Analysis

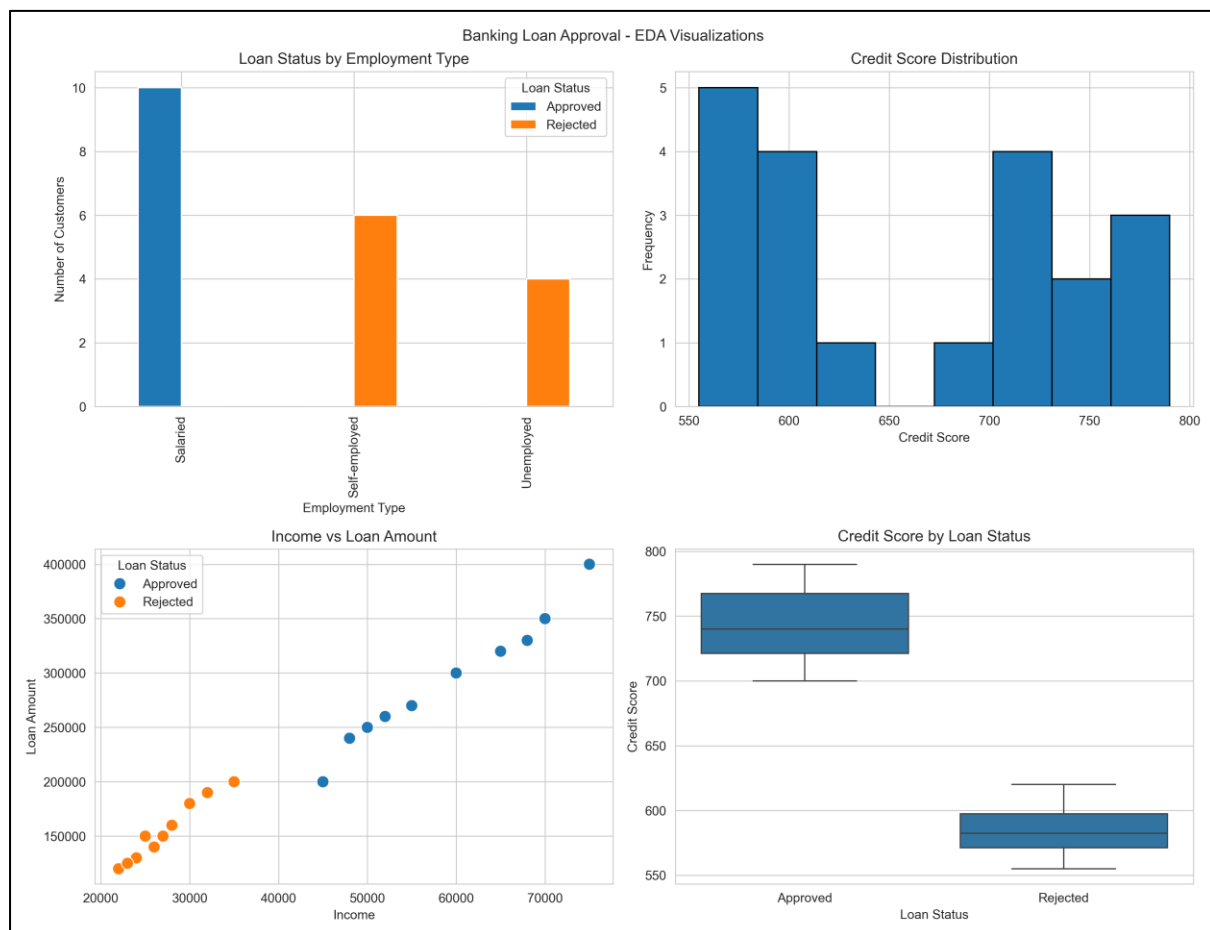
Employment Type

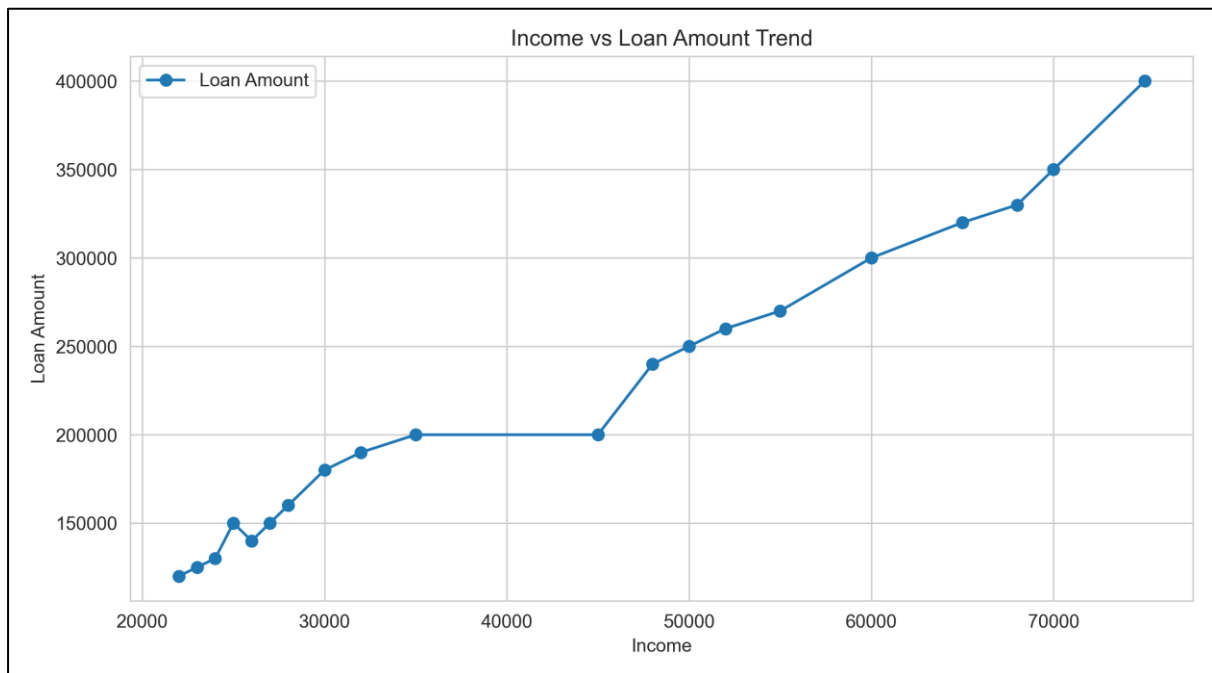
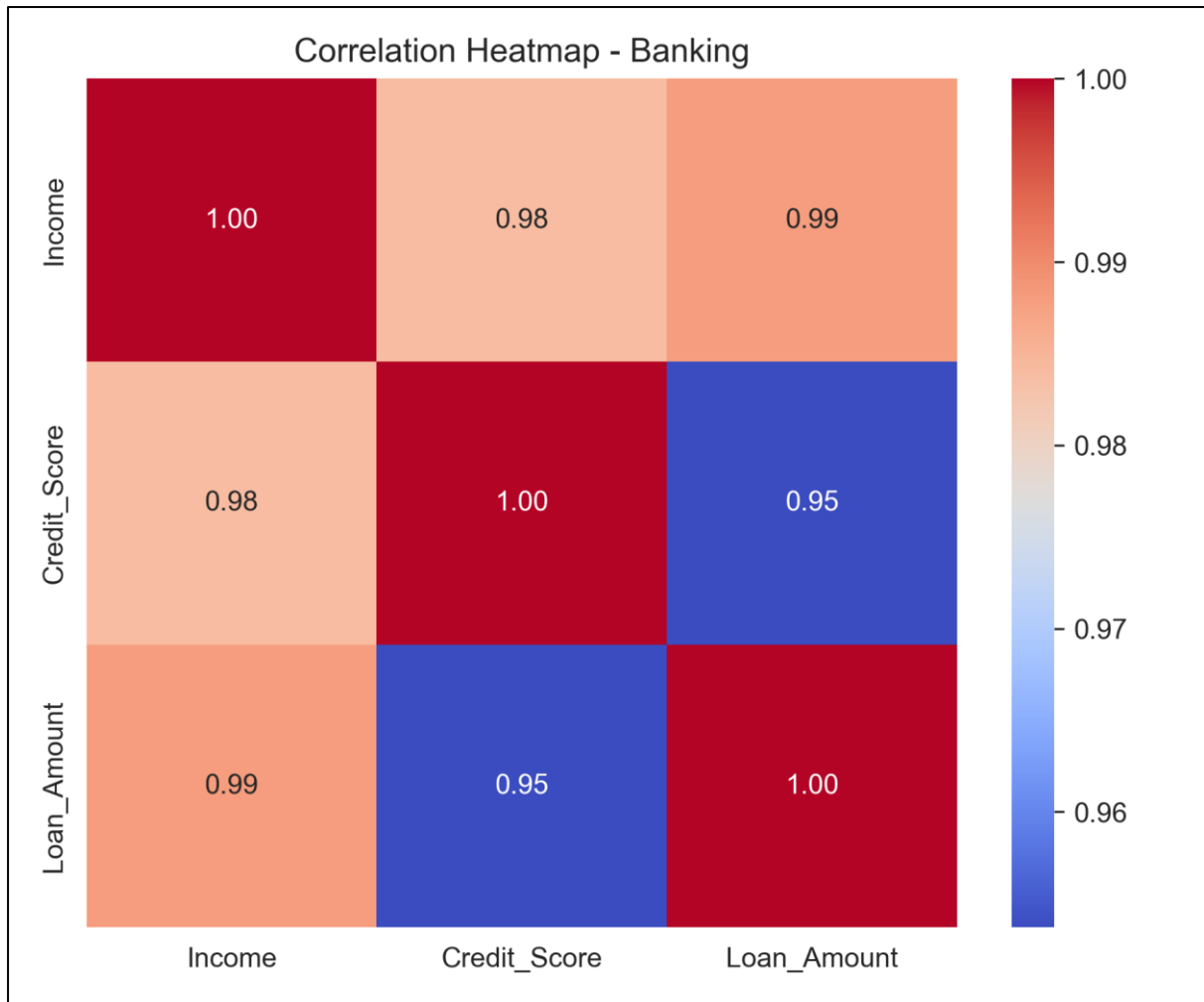
Category	Count	Percentage
Salaried	10	50%
Self-employed	6	30%
Unemployed	4	20%

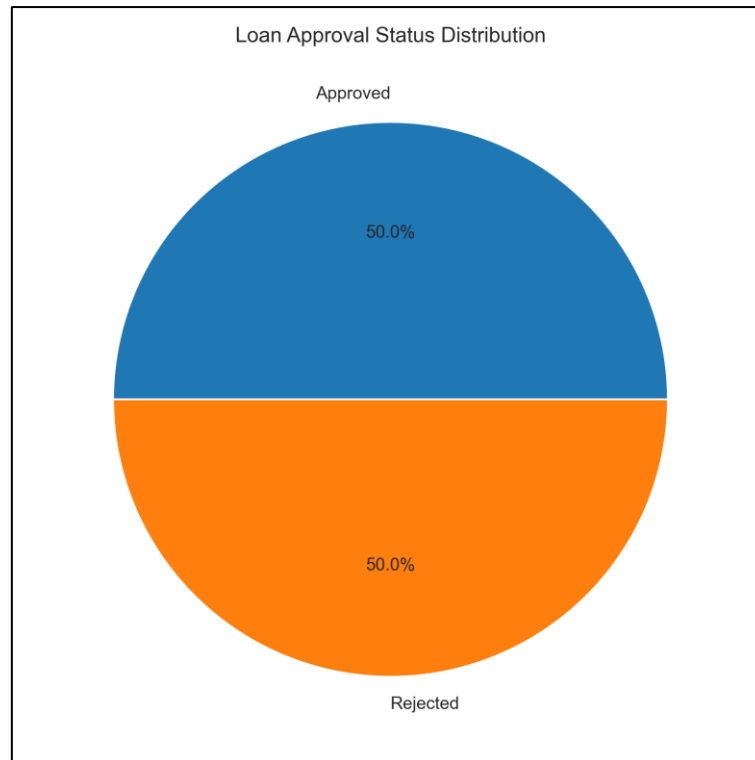
Loan Status

Category	Count	Percentage
Approved	10	50%
Rejected	10	50%

4.6 Visualizations







4.7 Trends, Distributions and Relationships

Approved applicants have credit scores between 700 and 790, while rejected applicants fall between 555 and 620, producing a clear separation. Income and Loan Amount are almost perfectly correlated ($r = 0.988$). All Salaried applicants were approved and all Unemployed applicants were rejected in this sample.

4.8 Key Findings and Insights

1. All Salaried applicants in this sample were approved.
2. All Unemployed applicants were rejected.
3. Approved applicants have credit scores between 700 and 790.
4. Higher-income customers request larger loan amounts.
5. No missing values, duplicates or IQR outliers were detected.

4.9 Decision-Making Impact

The credit-score gap and employment-type pattern can support a preliminary automated screening system, allowing clearly approvable or rejectable applications to be processed faster while human underwriters focus on borderline cases.

QUESTION 5 – AGRICULTURE CROP YIELD DATASET

5.1 Dataset Description

The Agriculture Crop Yield dataset contains **20 records and 6 features**. It includes Farm_ID, Rainfall_mm, Temperature, Fertilizer_kg, Soil_Type and Crop_Yield_kg.

Feature	Data Type	Role
Farm_ID	object	Unique identifier
Rainfall_mm	int64	Numerical predictor
Temperature	int64	Numerical predictor
Fertilizer_kg	int64	Numerical predictor
Soil_Type	object	Categorical
Crop_Yield_kg	int64	Numerical target

5.2 Purpose and Significance

An agricultural department wants to identify which factors most influence crop yield, including rainfall, temperature, fertilizer usage and soil type, to guide irrigation planning and soil-specific fertilizer recommendations.

5.3 Data Cleaning and Preprocessing

- 0 missing values.
- 0 duplicate rows.
- Soil_Type values were consistent.
- 0 IQR outliers were detected.

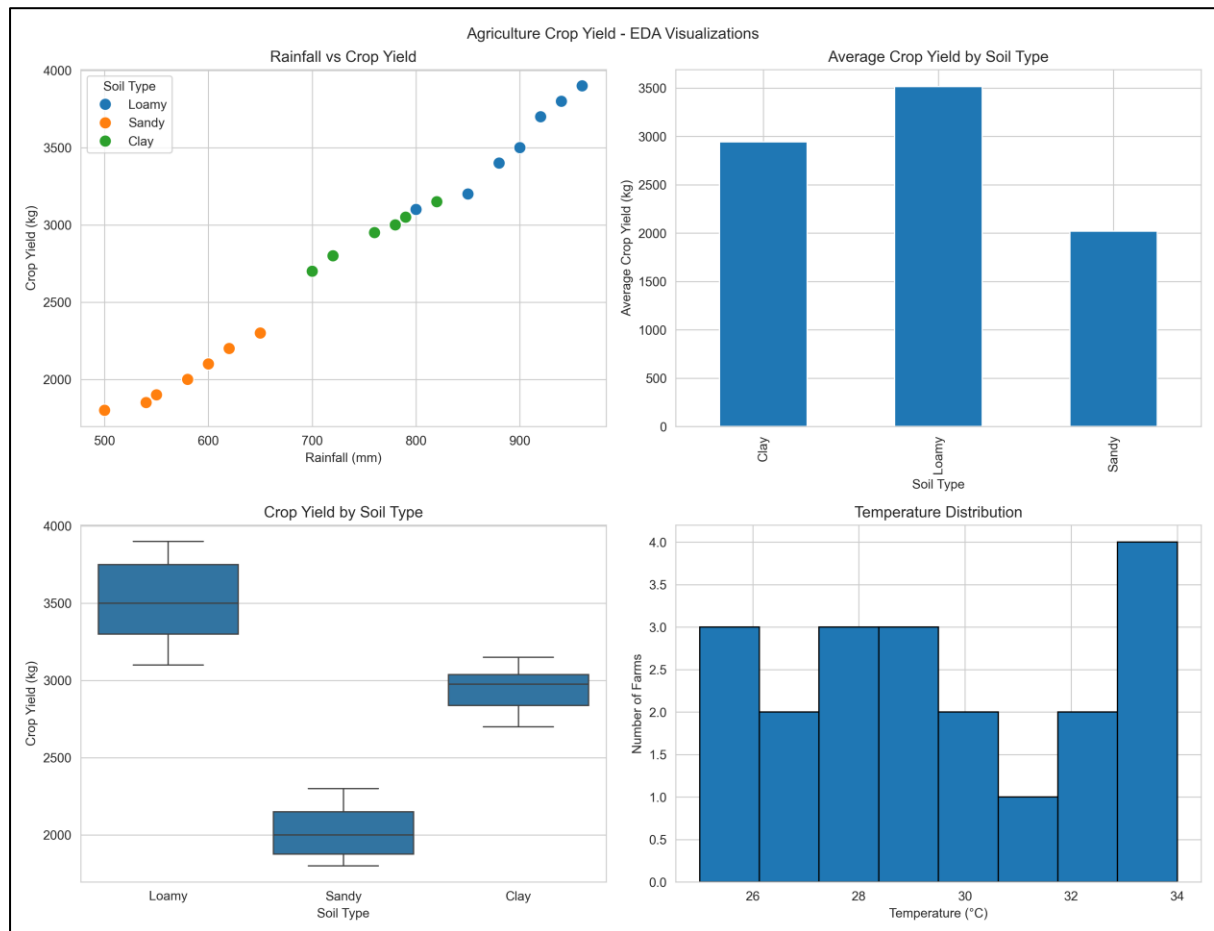
5.4 Descriptive Statistics

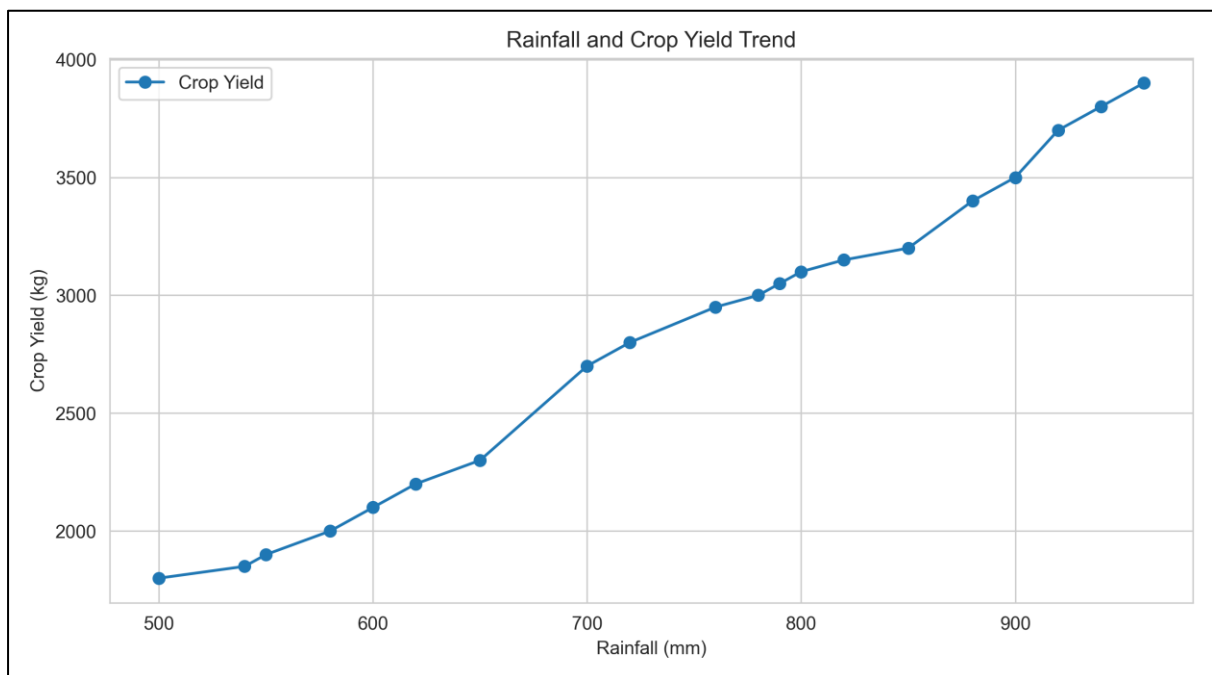
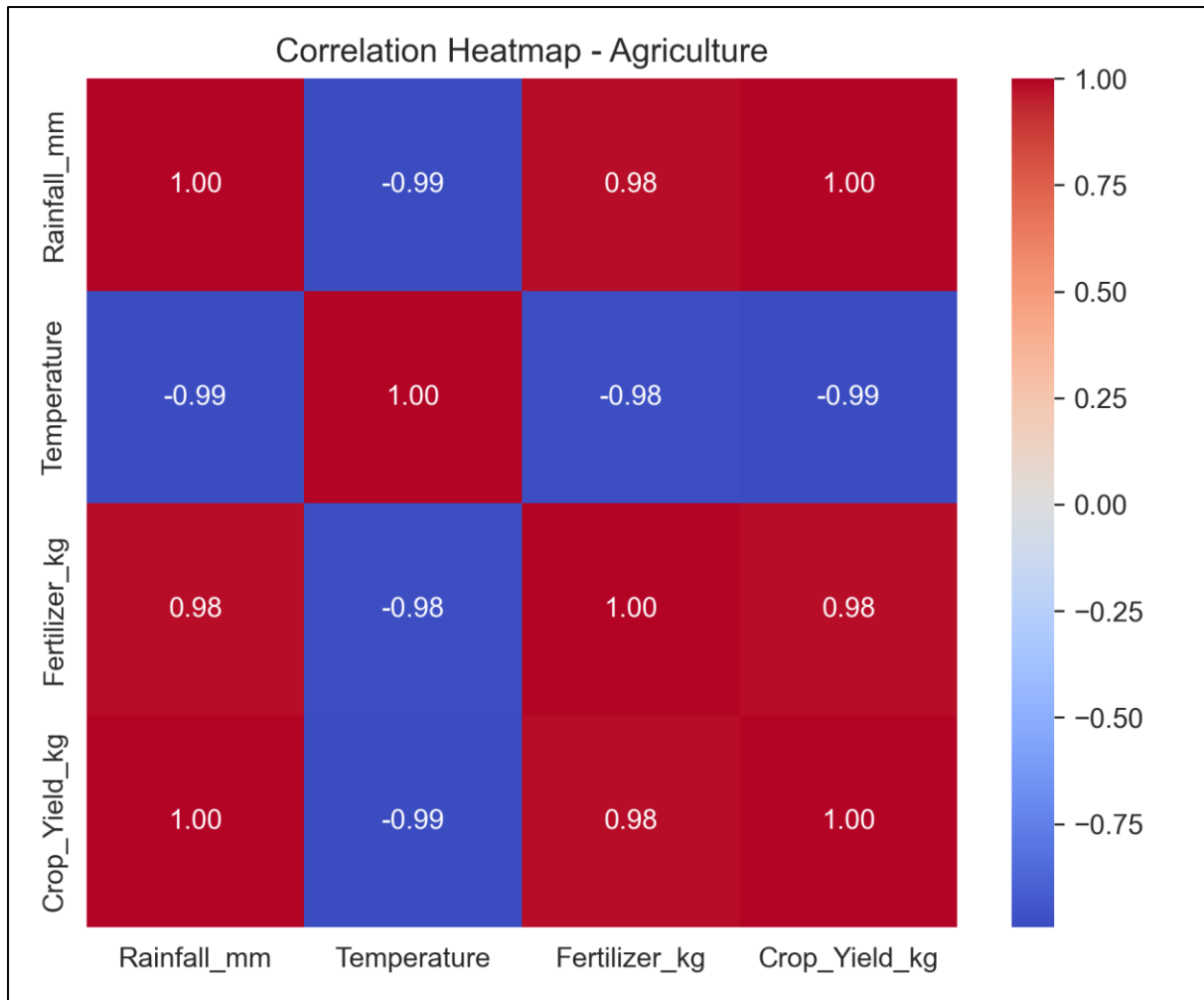
Feature	Mean	Maximum	Minimum	Std. Dev
Rainfall	743.00	960	500	144.34
Temperature	29.55	34	25	2.76
Fertilizer	54.75	75	40	10.51
Crop Yield	2820.00	3900	1800	681.41

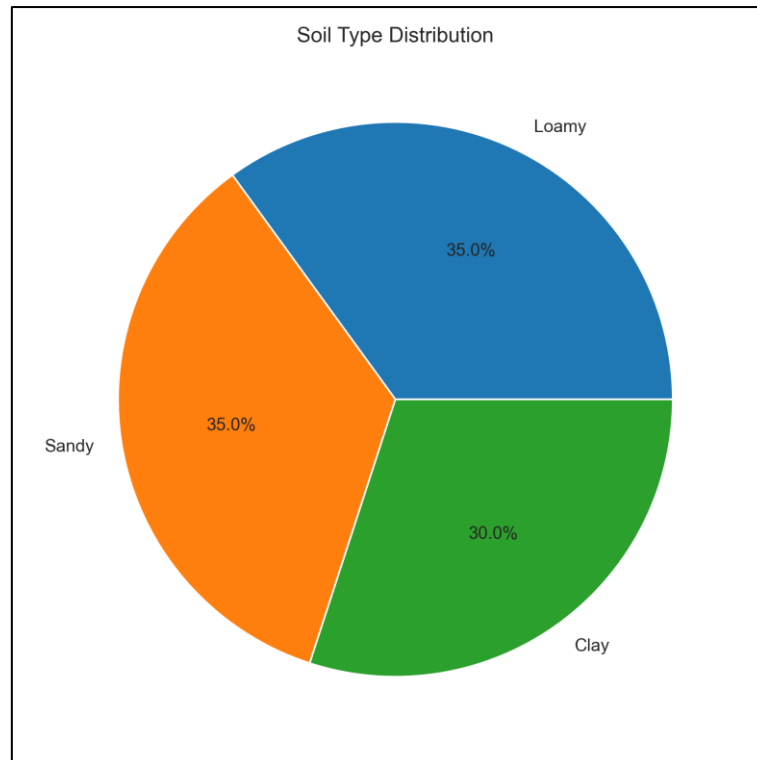
5.5 Categorical Variable Analysis

Soil Type	Count	Percentage
Loamy	7	35%
Sandy	7	35%
Clay	6	30%

5.6 Visualizations







5.7 Trends, Distributions and Relationships

Rainfall and Crop Yield have an almost perfect positive correlation ($r = 0.996$). Temperature is strongly negatively correlated with both rainfall and yield ($r = -0.993$). The line chart shows that crop yield follows the rainfall trend closely.

5.8 Key Findings and Insights

1. Rainfall has a very strong positive correlation with crop yield.
2. Loamy soil produces the highest average yield, followed by Clay and Sandy.
3. Temperature is strongly inversely related to crop yield.
4. Fertilizer usage has a positive correlation with crop yield ($r = 0.983$).
5. No missing values, duplicates or IQR outliers were detected.

5.9 Decision-Making Impact

The agricultural department can prioritize irrigation investment for low-rainfall Sandy-soil farms and combine irrigation improvements with fertilizer increases, which are positively associated with yield.